

Ian J Sawyer

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Computer Science Student Eager to Apply Knowledge in Low-Level Programming to Secure a Full-Time Position as a Low-Level Systems Software Engineer

Education

Bachelor of Science, Computer Science

May 2026

Rose-Hulman Institute of Technology, Terre Haute, IN

GPA: 2.76

Major GPA: 3.14

Relevant Courses: Computer Architecture, Software Design, Software Requirements Engineering, Operating Systems, Operating System Design, Network Security, Computer Networks, Design and Analysis of Algorithms

Skills

Programming Languages: C, Java, JavaScript, Python, SQL, bash

Tools & Technologies: Linux CLI, Git, Network programming (TCP/UDP, raw sockets), Network program debugging, MS Word, Object Oriented Design

Operating Systems: Linux, Windows

Work Experience

Montrow Group, North Vernon, IN

June 2025 - August 2025

CNC Wire Operator

- Collaborated with members of other departments to ensure timely and efficient job completion
- Designed efficient part set-ups for the mass-manufacturing of steel parts, improving workflow and efficiency.
- Performed detailed inspections using specification sheets and internal quality standards to ensure product accuracy and compliance
- Troubleshoot setup and production issues to minimize downtime and maintain output targets

Projects

Capstone Project: Puzzle Hunt Management Platform

August 2025 - Present

- Developed full-stack features for a puzzle hunt platform using MySQL, a Node.js REST API, and Svelte, enabling teams to securely access and solve puzzles during live events.
- Designed and implemented responsive user-facing interfaces in Svelte, improving usability for puzzle navigation, puzzle tracking, and live event interaction.

- Collaborated with a four-person engineering team and coordinated with our client to refine requirements and adjust implementations based on feedback.

Microkernel Operating System Research Project

December 2025 - Present

- Researched microkernel and monolithic kernel architectures to inform the design of a custom microkernel operating system, with emphasis on process isolation, inter-process communication (IPC), and kernel responsibility boundaries.
- Performed in-depth analysis of process control block (PCB) designs in existing production-level operating systems to design and implement a PCB, weighing real-world trade-offs in scheduling, fault handling, and capability management.
- Modernized the project's build infrastructure by migrating from a Make-based system to Meson, improving maintainability and iteration speed for development.

Personal Project: Network Stack for the Xv6 OS

June 2025 - August 2025

- Designed and implemented a custom network driver and UDP network stack for the Xv6 operating system, enabling inter-machine packet transmission.
- Built support for Ethernet, ARP, and UDP, as well as packet construction/parsing.
- Implemented a socket API into Xv6, modeled after Linux sockets, supporting `socket()`, `bind()`, `recvfrom()`, and `sendto()`.
- Gained hands-on experience debugging low-level networking in QEMU and with Wireshark, developing a deeper understanding of protocol interactions and driver/OS boundaries.
- Strengthened expertise in network protocol design (ARP, UDP, TCP fundamentals), concurrency control in device drivers, and the internals of Linux socket abstractions.

Activities

Rose-Hulman

- Competitive Programming Team
- Intramural Sports

2024 - 2025
2022 - Present

High-School

- Soccer - 2 years captain
- Baseball - 4 years varsity

2018 - 2022
2018 - 2022